

9	(a)	A satellite orbits the Earth of mass M in a circular path of radius r .	
		(i)	Show that the period T of the satellite is given by the expression $T^2 = \frac{4\pi^2}{GM} r^3$
			[3]
			Gravitational force F_G provides the centripetal force F_C. $F_C = F_G$ $mr\omega^2 = \frac{GMm}{r^2}$ $mr\left(\frac{2\pi}{T}\right)^2 = \frac{GMm}{r^2}$ $T^2 = \frac{4\pi^2}{GM} r^3$
		(ii)	A satellite is orbiting the Earth above the equator with a period of 28 hours. The mass of the Earth is 5.98×10^{24} kg.
			1. Calculate the radius of the satellite's orbit.
			radius = m [2]
			Using $T^2 = \frac{4\pi^2}{GM} r^3$ $r = \sqrt[3]{\frac{GM}{4\pi^2} (T^2)}$ $= \sqrt[3]{\frac{(6.67 \times 10^{-11})(5.98 \times 10^{24})}{4\pi^2} (28 \times 60 \times 60)^2}$ $= 4.68 \times 10^7 \text{ m}$

		2. The mass of the satellite is m. For the satellite in orbit, show that its kinetic energy E_K is given by $E_K = \frac{GMm}{2r}$	
			[2]
		$\frac{mv^2}{r} = \frac{GMm}{r^2}$ $E_K = \frac{1}{2}mv^2$ $= \frac{GMm}{2r}$	1 1
		3. Hence, determine the kinetic energy of the satellite if it has a mass of 1200 kg.	
		kinetic energy = J	[1]
		$E_K = \frac{GMm}{2r} = \frac{(6.67 \times 10^{-11})(5.98 \times 10^{24})(1200)}{2(4.68 \times 10^7)}$ $= 5.11 \times 10^9 \text{ J}$	1
		4. The satellite is then moved into a new orbit, gaining $1.14 \times 10^9 \text{ J}$ of gravitational potential energy in the process. Calculate the satellite's loss in kinetic energy.	
		loss in kinetic energy = J	[3]
		Given: Gain in gravitational potential energy $= -\frac{GMm}{r_2} - \left(-\frac{GMm}{r_1}\right) = 1.14 \times 10^9 \text{ J}$ where r_1 and r_2 are the radii of the old and new orbit respectively. Loss in kinetic energy $= KE_i - KE_f$ $= \frac{GMm}{2r_1} - \left(\frac{GMm}{2r_2}\right)$ $= \frac{1}{2} \left(\frac{GMm}{r_1} - \frac{GMm}{r_2} \right)$ $= \frac{1}{2} (1.14 \times 10^9)$ $= 5.70 \times 10^8 \text{ J.}$	1 1 1

- (b) A binary star consists of two stars that orbit about a fixed point C, as shown in Fig. 9.2.

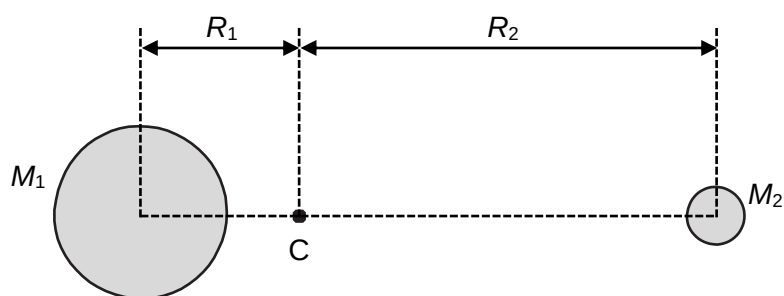


Fig. 9.2

The star of mass M_1 has a circular orbit of radius R_1 , and the star of mass M_2 has a circular orbit of radius R_2 . Rotating about point C, both stars have the same angular speed of $4.98 \times 10^{-8} \text{ rad s}^{-1}$.

- (i) Explain why the centripetal force acting on the two stars are equal in magnitude.

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..... [2]

The **gravitational force on each star exerted by the other star provides the centripetal force required** for each star's circular motion.

1

By Newton's 3rd law, the gravitational force that each star exerts on the other star are equal in magnitude and opposite in direction.

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- (ii) Calculate the period of orbit of each star.

period =years [2]

$$\omega = \frac{2\pi}{T}$$

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{4.98 \times 10^{-8}} \\ = 1.2617 \times 10^8 \text{ s} \\ = 4.00 \text{ years}$$

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- (iii) Show that the ratio of the masses of the stars is given by the expression

$$\frac{M_1}{M_2} = \frac{R_2}{R_1}$$

[1]

		Centripetal force experienced by both stars are equal, hence $M_1 R_1 \omega^2 = M_2 R_2 \omega^2$ Hence, $\frac{M_1}{M_2} = \frac{R_2}{R_1}$	1
	(iv)	Given that $\frac{M_1}{M_2} = 3.0$ and the separation between the stars is 3.2×10^{11} m, calculate the radius R_1 .	
		$R_1 = \dots\dots\dots$ m	[2]
		$\frac{M_1}{M_2} = \frac{R_2}{R_1} = 3.0$ $R_1 = \frac{1}{3.0}(R_2)$ Therefore, $R_1 = \frac{1}{4} \times (3.2 \times 10^{11})$ $= 8.0 \times 10^{10} \text{ m}$	1 1
	(v)	A planet orbits around the star of mass M_1 in the binary star system. Suggest why the orbit of the planet is not circular.	
			
			
			
			[2]
		When the planet and star of mass M_2 are on opposite sides of star of mass M_1, resultant force acting on the planet is large. When the planet and star of mass M_2 are on the same side of star of mass M_1, resultant force acting on the planet can be possibly be lower in magnitude. Hence, the orbit of the planet will not be a perfect circle. e.g. The star M_2 also exerts a gravitational force on the planet. As the planet orbits the star M_1, the resultant gravitational force due to both stars M_1 and M_2 will not remain constant as the relative positions between the stars and planet change during orbit. It will be the least when the planet is in between the two stars and greatest when M_1 is in between the planet and M_2.	1 1

			Since the resultant gravitational force on the planet provides the centripetal force for its orbit, the centripetal force is not constant and thus the orbit will not be circular.	

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